

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

ORDER NO. 92-047

WASTE DISCHARGE REQUIREMENTS
FOR
CITY OF TEHACHAPI
WASTEWATER TREATMENT FACILITY
KERN COUNTY

The California Regional Water Quality Control Board, Central Valley Region, (hereafter Board) finds that:

1. The City of Tehachapi (hereafter Discharger) submitted a Report of Waste Discharge, dated 19 September 1991, and a site evaluation report, dated August 1991, for an increase in the flow and for interim changes and improvements in the method of treatment and disposal of its wastewater.
2. The Discharger owns and operates an existing wastewater treatment facility (WWTF) that provides sewerage service to about 6000 residents and local business. The existing WWTF includes two Imhoff tanks (in parallel), a trickling filter, 14 evaporation/percolation ponds for effluent disposal, and a 140-acre parcel for reclamation purposes.
3. The ponds are adjacent to Tehachapi Creek. A storm drain channel, tributary to Tehachapi Creek, cuts across the WWTF and conveys storm water runoff from approximately 2800 acres north of Highway 58.
4. The discharge from the existing WWTF is predominantly domestic wastes, with some light industrial discharges, and is governed by Waste Discharge Requirements Order No. 91-153 and Cease and Desist Order No. 91-154, both adopted by the Board on 28 June 1991. Order No. 91-153 prescribes requirements for a discharge of 0.63 million gallons per day (mgd). Order No. 91-154 was adopted by the Board because of unauthorized discharges to Tehachapi Creek and violations of effluent limitations prescribed by waste discharge requirements.
5. The Discharger proposes to discharge a maximum of 0.85 mgd of domestic and light industrial wastes from an interim WWTF. The Discharger will abandon the Imhoff tanks and trickling filter and convert 7 of the existing disposal ponds into 2 aerated ponds and 5 stabilization ponds for treatment; continue using 6 of the existing evaporation/percolation ponds for disposal; use a 100 acre-feet storage reservoir to store treated wastewater during the winter; and continue reclamation in the 140-acre reclamation site cropped with alfalfa, as weather permits. The changes and improvements described in this Finding comprise the interim WWTF.
6. The Discharger proposes to build a new 1.25-mgd WWTF and contract for reclamation areas as a long-term solution to dispose of increases in flow. The new 1.25-mgd WWTF will include a control building,

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headworks, a 1.25-mgd activated sludge oxidation ditch, a secondary clarifier, about 4000 square feet of sludge drying beds, the 100 acre-feet storage reservoir, and the 140-acre reclamation site. The Discharger completed preliminary plans and specifications for the new WWTF in 1990, began construction of the headworks in August 1991, and plans to continue construction of the rest of the new WWTF by July 1992 and complete it in August 1984.

7. The Discharger proposes to use sludge from the interim and new WWTFs as a soil amendment.
8. The 140-acre reclamation area is adjacent to the Tehachapi Kern County Airport No. 4, about 125 feet from the main runway. A residential tract is about 500 feet east of the reclamation area. Soils at the reclamation area are classified as sandy silts, silty sands, and clays, with permeabilities varying from 9×10^{-9} to 6×10^{-6} cm/sec.
9. The Discharger owns the 100 acre-feet storage reservoir. The reservoir is about 2 1/2 miles southeast of the existing WWTF and is surrounded by predominantly agricultural land. The reservoir is about 30 feet deep and has a bottom area of 10 acres. The Discharger proposes to build a 10-foot high berm across the reservoir to divide it into two cells with a combined storage capacity of 100 acre-feet. Based on three borings conducted in 1990, soils beneath the reservoir were classified as clayey silt and silty clay with low permeabilities. Additional soil sampling is being conducted to verify the types of soils beneath the reservoir and to make appropriate recommendations to improve reservoir storage capacity.
10. The Discharger is on the State Water Resources Control Board's 1991-92 Revolving Fund Priority List for consideration of a \$5,237,000 low interest loan to maximize reclamation.
11. Data in the August 1991 technical report indicates that the proposed interim improvements and modifications will allow the Discharger to treat, store, and dispose of 0.85 mgd. However, the Discharger has not substantiated that it has disposal capacity for 1.25 mgd.
12. The existing and interim WWTFs are in Section 20, T32S, R33E; the proposed new WWTF will be in Section 20, T32S, R33E; the 100 acre-feet storage reservoir is in Section 22, T32S, R14W, and the 140-acre alfalfa field is in Section 21, T32S, R33E, MDB&M, with surface water drainage to Tehachapi Creek, as shown in Attachment A, which is attached hereto and part of the Order by reference. The site lies within the Grapevine Hydrologic Unit (No. 556.10), as depicted on interagency hydrologic maps prepared by the Department of Water Resources in August 1986.
13. The Board adopted a Water Quality Control Plan for the Tulare Lake Basin (hereafter Basin Plan), which contains water quality objectives

for all waters of the Basin. These requirements implement the Basin Plan.

14. The Basin Plan states that the minimum treatment level for WWTFs which discharge more than 1.0 mgd is secondary treatment, which is defined in the Basin Plan as 80 percent removal of BOD₅ and suspended solids, or reduction to 40 mg/l, whichever is more restrictive.
15. Tehachapi Creek is an ephemeral Westside stream. It originates in the Tehachapi Valley, about two miles south of the City. Tehachapi Valley has a 39,700-acre watershed area. Kern County Water Agency reports that flows in Tehachapi Creek reached a maximum of 33.7 cubic feet per second in December of 1988. The stream exits the Valley at an elevation of 3900 feet (MSL), cuts across Keene Ranch, and joins Caliente Creek one-half mile west of the town of Caliente. Caliente Creek is also an ephemeral stream that drains to the valley floor.
16. The beneficial uses of Tehachapi Creek are agricultural use, wildlife habitat, preservation of rare and endangered species, noncontact recreation (at Keene Ranch), and ground water recharge.
17. Ground water beneath the existing WWTF is 7 to 20 feet below ground surface (bgs) and moves generally to west. Ground water below the 140-acre reclamation area is about 80 to 120 feet bgs and moves to the southwest. Ground water beneath the storage reservoir is 130 to 170 feet bgs and moves to the west. There are no wells within 500 feet of the WWTF, the reclamation area, or storage reservoir, but there is a domestic well about 750 feet from the reservoir. Based on water samples collected in 1991 from the City's domestic wells, ground water is generally of good to excellent quality, with an EC of 369 to 625 μ mhos/cm, nitrates of 0 to 4 mg/l, total dissolved solids of 300 to 430 mg/l, and pH of 7.8.
18. The beneficial uses of ground water in the Tehachapi Valley are domestic, industrial, and agricultural supply.
19. Ground water beneath a 23-acre parcel adjacent to the ponds is not fit for domestic uses, and may have very limited agricultural or industrial uses. Mr. John Broom owns the 23-acre parcel and granted the City permission to use the parcel for storage, containment, and disposal of winter overflows from December 1990 to April 1991.
20. The Discharger submitted a revised technical report dated 10 December 1991 and entitled *Workplan for Groundwater Monitoring Network Tehachapi Wastewater Treatment Facility, Tehachapi, California* outlining a ground water monitoring network consisting of one ground water monitoring well upgradient of the WWTF and three ground water monitoring wells downgradient of the WWTF and the 23-acre parcel to monitor the effects of the discharge on ground water quality.

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21. The State Department of Health Services has promulgated statewide criteria for the uses of reclaimed water in Title 22, California Code of Regulations (CCR), Section 60301, et seq., hereafter Title 22.
22. On 14 June 1991, the City of Tehachapi certified a final environmental impact report (EIR) in accordance with the California Environmental Quality Act (CEQA) (Public Resources Code Section 21000, et seq.) and the State CEQA Guidelines.
23. The Board has reviewed the EIR and finds that compliance with these requirements will mitigate and prevent any significant impacts on water quality.
24. The Board has notified the Discharger and interested agencies and persons of its intent to prescribe waste discharge requirements for this discharge and has provided them with an opportunity for a public hearing and an opportunity to submit their written views and recommendations.
25. The Board, in a public meeting, heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that Order No. 91-153 is rescinded and that the City of Tehachapi, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereunder, shall comply with the following:

A. Discharge Prohibitions:

1. Discharge of wastes to surface waters or surface water drainage courses is prohibited.
2. Bypass or overflow of untreated or partially treated waste is prohibited.
3. Discharge of waste classified as 'hazardous' or 'designated', as defined in Sections 2521(a) and 2522(a) of Chapter 15, is prohibited.

B. Discharge Specifications:

1. Subject to satisfaction of Provision F.13, the monthly average daily dry weather discharge flow shall not exceed 1.25 million gallons.

2. Effluent shall not exceed the following limits:

<u>Constituent</u> ^{1/}	<u>Units</u>	<u>Monthly Average</u>	<u>Daily Maximum</u>
BOD ₅ ^{2/}	mg/l	40	80
CBOD ₅ ^{3/}	mg/l	35	70
Total Suspended Solids	mg/l	40	80
Settleable Solids	ml/l	0.2	0.5

- ^{1/} The Discharger must demonstrate compliance with either BOD₅ or CBOD₅ effluent limitations.
- ^{2/} Five-day, 20° Celsius biochemical oxygen demand.
- ^{3/} Five-day, 20° Celsius carbonaceous biochemical oxygen demand.
3. Ponds and impoundments shall not have a pH less than 6.5 or greater than 9.5.
4. Objectionable odors originating at the treatment, storage, disposal, and reclamation facilities shall not be perceivable beyond the limits of the treatment, storage, disposal, and reclamation facilities.
5. As a means of discerning compliance with Discharge Specification No. B.4, the dissolved oxygen content in the upper zone (1 foot) of wastewater in ponds and impoundments shall not be less than 1.0 mg/l.
6. The treatment facilities shall be designed, constructed, operated, and maintained to prevent inundation or washout due to floods with a 100-year return frequency.
7. The maximum specific electrical conductivity (EC) of the discharge shall not exceed the average EC of the source water plus 500 μ nhos/cm, or a maximum of 900 μ nhos/cm, whichever is less.
8. Ponds and impoundments shall be managed to prevent breeding of mosquitos. In particular:
- An erosion control program should assure that small coves and irregularities are not created around the perimeter of the water surface.
 - Weeds shall be minimized through control of water depth, harvesting, or herbicides.

- c. Dead algae, vegetation, and debris shall not accumulate on the water surface.
- 9. Public contact with wastewater shall be precluded through such means as fences, signs, and other acceptable alternatives.
- 10. Ponds and impoundments shall have sufficient capacity to accommodate allowable wastewater flow and design seasonal precipitation and ancillary inflow and infiltration during the nonirrigation season. Design seasonal precipitation shall be based on total annual precipitation using a return period of 100 years, distributed monthly in accordance with historical rainfall patterns. Freeboard shall never be less than two feet (measured vertically).
- 11. On or about 1 October of each year, available storage capacity shall at least equal the volume necessary to comply with Discharge Specification B.10.

C. Reclamation Specifications:

- 1. Use of reclaimed water shall be limited to fodder, fiber, and seed crops and construction purposes and shall comply with the appropriate criteria contained in Section 60309 of Title 22.
- 2. Public contact with reclaimed water shall be precluded through such means as fences, signs, and irrigation management practices.
- 3. Areas irrigated with reclaimed water shall be managed to prevent breeding of mosquitos. More specifically:
 - a. Tail water must be returned and all applied irrigation water must infiltrate completely within a 24-hour period.
 - b. Ditches not serving as wildlife habitat should be maintained free of emergent, marginal, and floating vegetation.
 - c. Low-pressure and unpressurized pipelines and ditches accessible to mosquitos shall not be used to store reclaimed water.
- 4. Reclaimed water used for irrigation shall be managed to minimize erosion, runoff, and movement of aerosols from the disposal area.
- 5. Application of reclaimed water to the reclamation area shall not exceed the water demands or nitrate uptake of the crop, including contribution of other sources of nitrates applied to the area.

6. The Discharger may not irrigate the reclamation area with effluent during periods of precipitation and for at least 24 hours after cessation of precipitation.
7. Irrigation or impoundment of wastewater shall not occur within 500 feet of any domestic well or within 100 feet of any irrigation well unless it is demonstrated to the satisfaction of the Executive Officer that a lesser distance is justified.
8. Storm water runoff from the reclamation area shall not be discharged to any surface water drainage course within 30 days of the last wastewater application.

D. Sludge Disposal:

1. Collected screenings, sludges, and other solids removed from liquid wastes shall be disposed of in a manner that is consistent with Chapter 15 and approved by the Executive Officer.
2. Any proposed change in sludge use or disposal practice shall be reported to the Executive Officer at least 90 days in advance of the change.
3. Use and disposal of sewage sludge shall comply with existing federal and state laws and regulations, and with Clean Water Act (CWA) Section 405(d) technical standards when promulgated.

If an applicable management practice or numerical limitation for pollutants in sewage sludge is promulgated under Section 405(d) of the CWA after issuance of this permit which is more stringent than the sludge pollutant limit or management practice specified in this permit or in existing federal or state laws or regulations, the Discharger shall comply with the limitations by not later than the compliance deadline specified in the applicable regulations as required by Section 405(d) of the CWA.

E. Ground Water Limitations:

The discharge, in combination with other sources, shall not cause underlying ground water to:

1. Contain waste constituents in concentrations statistically greater than receiving water limits, where specified below, or background water quality where not specified. (For purposes of comparison, background water quality shall be determined when background

monitoring provides sufficient data. Quality determined in this manner establishes "water quality protection standards.")

2. Exceed an annual average incremental increase in specific electrical conductivity greater than 3 μ mhos/cm, based on the most recent five-year period, or a maximum of 900 μ mhos/cm., whichever is less.
3. Contain chemicals, heavy metals, or trace elements in concentrations that adversely affect beneficial uses or exceed maximum contaminant levels specified in 22 CCR, Division 4, Chapter 15.
4. Exceed a most probable number of total coliform organisms of 2.2/100 ml over any seven-day period.
5. Exceed concentrations of radionuclides specified in 22 CCR, Division 4, Chapter 15.
6. Contain taste or odor-producing substances in concentrations that cause nuisance or adversely affect beneficial uses.
7. Contain concentrations of chemical constituents in amounts that adversely affect agricultural use.

F. Provisions:

1. The Discharger shall comply with the Monitoring and Reporting Program No. 92-047, which is part of this Order, and any revisions thereto as ordered by the Executive Officer.
2. The Discharger shall comply with the "Standard Provisions and Reporting Requirements for Waste Discharge Requirements", dated 1 March 1991, which are attached hereto and by reference a part of this Order. This attachment and its individual paragraphs are commonly referenced as "Standard Provision(s)."
3. In the event of any change in control or ownership of land or waste discharge facilities described herein, the Discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be immediately forwarded to this office.
4. At least 120 days prior to termination or expiration of any lease, contract, or agreement involving disposal or reclamation areas or off-site reuse of effluent, if used to justify the capacity authorized herein and assure compliance with this Order, the Discharger shall notify the Board in writing of the situation and of what measures have been taken or are being taken to assure full compliance with this Order.

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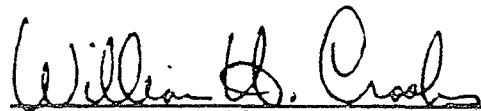
5. The Discharger shall use the best practicable cost-effective control technique currently available to comply with salinity limits specified in this order.
6. The Discharger must comply with all conditions of this Order, including timely submittal of technical and monitoring reports as directed by the Executive Officer. Violations may result in enforcement action, including Regional Board or court orders requiring corrective action or imposing civil monetary liability, or in revision or rescission of this Order.
7. To accept septage wastes or wastes other than domestic wastes at the interim or new WWTF, the Discharger shall submit an engineering report demonstrating that the WWTF can adequately control the quality and quantity of said wastes without interfering with the normal performance and operation of the existing plant. The report is subject to the approval of the Executive Officer, shall be submitted at least sixty days prior to accepting said wastes at the plant, and shall be prepared by a California registered civil engineer experienced in wastewater treatment and disposal.
8. Prior to 15 May 1992, the Discharger shall submit to the Board a final irrigation management plan for the use of reclaimed water at the 140-acre alfalfa reclamation site. The irrigation management plan shall include a water balance and a nitrogen balance for current and projected flows, with storage requirements using annual precipitation data with a 100-year return frequency, and must demonstrate that reclamation can be accomplished in accordance with accepted irrigation practices without contributing additional nitrogen in the form of nitrate ion to the ground water. The plan shall also address compliance with Discharge Specification B.11 and Reclamation Specifications C.2 to C.5, and shall be prepared by a California registered civil engineer experienced in wastewater treatment and disposal.
9. The Discharger shall submit to the Board on or before each compliance report due date, the document itself if so specified or a written report detailing compliance or noncompliance with the specific schedule date and task. If noncompliance is being reported, the reasons for such noncompliance shall be stated, plus an estimate of the date when the Discharger will be in compliance. The Discharger shall notify the Board by letter when it returns to compliance with the time schedule.
10. A copy of this Order shall be kept at the discharge facility for reference by operating personnel. Key operating personnel shall be familiar with its contents.

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11. If reclaimed water is used for construction purposes, it shall comply with the most current edition of "Guidelines for Use of Reclaimed Water for Construction Purposes". Other uses of reclaimed water not specifically authorized herein shall be subject to the approval of the Executive Officer and shall comply with 22 CCR, Division 4.
12. The Board will review this Order periodically and will revise requirements when necessary.
13. Until the new WWTF is complete and fully operational, and a registered civil engineer certifies in writing that it has the capacity, as designed and constructed, to treat and dispose of average dry weather flow of 1.25 mgd, and associated daily peak and wet weather flows, and consistently comply with this Order, monthly average dry weather flow shall be limited to 0.85 mgd.

I, WILLIAM H. CROOKS, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 27 March 1992.


WILLIAM H. CROOKS, Executive Officer

JA:cjs: 3/27/92

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
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INFLUENT MONITORING

Samples shall be collected at approximately the same time as effluent samples and should be representative of the influent for the period sampled. Influent monitoring shall include at least the following:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>
Settleable Solids	ml/l	Grab	Weekly
20° C <u>BOD₅</u>	mg/l	8-hr Composite	Weekly
20° C <u>CBOD₅</u>	mg/l	8-hr Composite	Weekly
Suspended Solids ^{1/}	mg/l	8-hr Composite	Weekly

^{1/} Monitoring for this constituent shall be implemented when flows reach or exceed 0.85 mgd or when the new 1.25-mgd WWTF is completed, whichever becomes first.

EFFLUENT MONITORING

Effluent samples shall be collected just prior to discharge to the disposal facility and should be representative of the volume and nature of the discharge. For the interim WWTF, samples collected from the outlet structure of ponds or the storage cells will be considered adequately composited. Time of collection of a grab sample shall be recorded. Effluent monitoring shall include at least the following:

<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency^{1/}</u>
Flow	mgd	Metered	Daily
Settleable Solids	ml/l	Grab	Daily
Specific Conductivity	μmhos/cm	Grab	Weekly
pH	pH Units	Grab	Weekly
20° C BOD ₅	mg/l	8-hr composite	Weekly
20° C CBOD ₅	mg/l	8-hr composite	Weekly
Suspended Solids ^{2/}	mg/l	8-hr composite	Weekly

(see footnotes on next page).

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- 1/ If results of monitoring a pollutant appear to violate effluent limitations, but monitoring frequency is not sufficient to validate violation (e.g., the monthly mean for BOD), or indicate a violation and potential upset of the treatment process (e.g., less than minimum D.O.), the frequency of sampling shall be increased to confirm the magnitude and duration of violation, if any, and aid in identification and resolution of the problem.
- 2/ Monitoring for this constituent shall be implemented when flows reach or exceed 0.85 mgd or when the new 1.25-mgd WWTF is completed, whichever becomes first.

POND AND STORAGE RESERVOIR MONITORING

The freeboard shall be monitored on all disposal ponds and the cells of the storage reservoir in use to the nearest tenth of a foot. Pond and storage reservoir monitoring shall include at least the following:

<u>Constituent</u>	<u>Unit</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>
Dissolved Oxygen ^{1/}	mg/l	Grab	Daily
Freeboard	feet	Observation	Daily

- 1/ Samples shall be collected at a depth of one foot from each pond and cell of the storage reservoir, opposite the inlet, and analyzed for dissolved oxygen. Samples shall be collected between 0800 and 0900 hours.

Permanent markers shall be placed in each pond and cell of the reservoir with calibrations indicating the water level at design capacity and available operational freeboard.

In addition, the Discharger shall inspect the condition of the ponds and the cells of the reservoir once per week and write visual observations in a bound log book. Notations shall include observations of whether weeds are developing in the water or along the bank, and their location; whether dead algae, vegetation, scum, or debris are accumulating on the pond surface and their location; whether burrowing animals or insects are present; and the color of the pond/cell (e.g., dark sparkling green, dull green, yellow, grey, tan, brown, etc.). A copy of the entries made in the log during each month shall be submitted along with the monitoring report the following month. Where the O&M manual indicates remedial action is necessary, the Discharger shall briefly explain in the transmittal what action has been taken or is scheduled to be taken.

WATER SUPPLY MONITORING

A sampling station shall be established where a representative sample of the water supply can be obtained. Water supply monitoring shall include at least the following:

<u>Constituents</u> ^{1/}	<u>Units</u>	<u>Sampling Frequency</u>
Standard Minerals	mg/l	Annually
Total Dissolved Solids	mg/l	Annually
Specific Conductivity @ 25°C	µmhos/cm	Monthly

- ^{1/} Constituents shall be reported as a flow weighted average from all wells and report shall include copies of supporting calculations.

RECLAMATION AREA MONITORING

The reclamation area monitoring shall be implemented by the Discharger no later 30 November 1992. A minimum of 5 representative locations shall be established, subject to the review and approval of the Executive Officer, for soil profile sampling at the reclamation area. Monitoring of the reclamation area shall at least include the following:

<u>Measurement</u>	<u>Unit</u>	<u>Soil Profile</u>	<u>Sampling Frequency</u>
Nitrate Nitrogen	mg/kg	10 feet ^{1/}	Annually ^{2/}
Kjeldahl Nitrogen	mg/kg	10 feet ^{1/}	Annually ^{2/}
Total Salts	mg/kg	10 feet ^{1/}	Annually ^{2/}

- ^{1/} Samples shall be collected at 1-foot increments within the root zone and 5-foot increments thereafter.
- ^{2/} Each location shall be sampled at the end of the growing season.

The Discharger shall inspect the designated reclamation area weekly. During the period of 1 November through 30 April, the Discharger shall inspect the area daily. In conducting the inspection, a log shall be kept of the condition of the area. Special attention shall be given to compliance with Discharge Specification B.11. Attention shall also be given to the presence or absence of the following:

- | | |
|----------------------|-------------------|
| a. Breaches of berms | c. Erosion |
| b. Vectors | d. Standing water |

A summary, addressing the items a through d listed above, shall be included with monthly monitoring report.

GROUND WATER MONITORING

By 30 May 1992, the Discharger shall installed the ground water monitoring network described in Finding No. 20 of this Order consisting of one background ground water monitoring well and three downgradient ground water monitoring wells in the area of the disposal ponds and the adjacent 23-acre parcel.

The downgradient wells shall constitute "points of compliance (POCs)". In conjunction with background monitoring, monitoring of POCs will enable one to determine compliance with water quality protection standards. The ground water surface elevation (in feet and hundredths, M.S.L.) in all wells shall be measured on a quarterly basis and used to determine the gradient and direction of ground water flow. This information shall be displayed on a water flow net diagram for the site. Water samples shall be collected from wells in the approved monitoring network and analyzed as follows:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>
Specific Electrical Conductance	μ mhos/cm	Grab	Quarterly
pH	pH Units	Grab	Quarterly
Standard Minerals ^{1/}	mg/l	Grab	Quarterly

^{1/} Standard mineral analyses shall include bicarbonate, calcium, carbonate, chloride, fluoride, iron, magnesium, nitrate, potassium, sodium, sulfate, total dissolved solids, and total phosphorous.

If the Discharger, through a verification monitoring program, or the Board verifies that water quality protection standards have been exceeded at or beyond the POC's, the Discharger shall notify the Board, or acknowledge the Board's findings, and submit a technical report within 90 days. The report must contain a plan and time schedule for implementing a corrective action program designed to achieve compliance with water quality protection standards.

WATER QUALITY PROTECTION STANDARDS

Standards for Ground Water

Monthly samples shall be collected from background monitoring wells for one year and analyzed for the parameters specified in "Ground Water Monitoring", above. Data from these analyses shall be reported to the Board by 15 June 1993 for use in determining water quality protection standards for ground water at the disposal site.

If subsequent sampling of background monitoring wells indicates significant water quality changes due to either seasonal fluctuations or other reasons unrelated to waste management activities at the facility, the Discharger may request modifications of these water quality protection standards.

SLUDGE MONITORING

When sludge is removed from ponds and treatment units, but prior to disposal, a composite sample shall be analyzed, on a dry weight basis, for Total Solids (%), Nitrogen (total, $\text{NH}_4\text{-N}$, and $\text{NO}_3\text{-N}$), Total Phosphorous, Total Potassium, Total PCBs, and totals of specific metals (Pb, Zn, Cu, Ni, Cd, and Ag). Analytical results shall be submitted to the Executive Officer. Analysis of soluble concentrations of these specific metals shall also be included as needed. If final disposal is proposed to go to land, a technical report analyzing application rates and procedures relative to Department of Health Services' *Manual of Good Practices for Landspreading of Sewage Sludge* and EPA's *Process Design Manual for Land Application of Municipal Sludges* and Title 23, California Code of Regulations, Section 2511(f), shall be completed and submitted to the Executive Officer for approval. The report shall be prepared by a California registered civil engineer experienced in wastewater treatment and disposal.

REPORTING

Monthly monitoring reports shall be submitted to the Regional Board by the 15th day of the following month. Quarterly reports shall be submitted to the Regional Board by 15 April, 15 July, 15 October, and 15 January of each year.

In reporting the monitoring data, the Discharger shall arrange the data in tabular form so that the date, the constituents, and the concentrations are readily discernible. The data shall be summarized in a manner that illustrates clearly whether the Discharger complies with waste discharge requirements, including calculation of all averages, etc.

If the Discharger monitors any pollutant at the locations designated herein more frequently than is required by this Order, the results of such monitoring shall be included in the discharge monitoring report.

The Discharger may also be requested to submit an annual report to the Board with tabular and graphical summaries of the monitoring data obtained during the previous year. Any such request shall be made in writing. The report shall discuss the corrective actions taken and planned to bring the discharge into full compliance with the waste discharge requirements.

By 31 January of each year, the Discharger shall submit a written report to the Executive Officer containing the following:

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- a. The names, certificate grade and general responsibilities of persons operating and maintaining the wastewater treatment plant (Standard Provision E.1).
- b. The names and telephone numbers of persons to contact regarding plant emergency and routine situations.
- c. A statement certifying when the flow meter and other monitoring instruments and devices were last calibrated, including identification of who performed the calibration (Standard Provision C.4).
- d. A statement identifying whether the current operation and maintenance manual, and contingency plan, reflect the wastewater treatment plant as currently constructed and operated, and the dates when these documents were last reviewed for adequacy.
- e. The total quantity of sludge disposed of during the previous year and ultimate disposal site(s).
- f. A summary of the reclamation area operation discussing total water application over the season; the total wastewater applied; the total nutrient loading from wastewater, sludges, and chemical fertilizers; and amount of nutrients removed through harvest of the crop. In short, the report shall present a mass balance relative to pollutants of concern and hydraulic loading.

All reports submitted in response to this Order shall comply with the signatory requirements in Standard Provision B.3.

The Discharger shall implement this monitoring program on the first day of the month following adoption of this Order.



WILLIAM H. CROOKS, Executive Officer

27 March 1992
(Date)

JA:cjs: 3/27/92

INFORMATION SHEET

CITY OF TEHACHAPI WWTF KERN COUNTY

The City of Tehachapi (Discharger) proposes to build a 1.25-mgd wastewater treatment facility (WWTF) as a long-term solution to handle increases in flow. It submitted a Report of Waste Discharge, dated 19 September 1991, and a technical report in August 1991 that outline an interim wastewater management plan for a discharge of 0.85 million gallons per day (mgd) of predominantly domestic wastes, with some light industrial discharges. This tentative Order prescribes waste full secondary effluent limits for the new WWTF and implements guidelines promulgated in the *"Water Quality Control Plan for the Tulare Lake Basin"* (5D).

The Discharger is currently governed by Waste Discharge Requirements Order No. 91-153 and Cease and Desist Order (C&D) No. 91-154, both adopted by the Board on 28 June 1991. Order No. 91-153 prescribes requirements for a discharge of 0.63 mgd. Order No. 91-154 was adopted by the Board because of unauthorized discharges to Tehachapi Creek and violations of effluent limitations prescribed by waste discharge requirements. Along with these tentative requirements, staff is recommending that the Board consider an amendment to C&D Order No. 91-154. The proposed amended C&D Order prescribes interim effluent limits for a discharge of 0.85 mgd and provides a compliance schedule for an increase in flow to 1.25 mgd. These issues are detailed in the staff report accompanying the proposed amendment.

The existing WWTF provides sewerage service to about 6000 residents and local businesses and includes dual Imhoff tanks, a trickling filter, fourteen evaporation/percolation ponds, and a 140-acre alfalfa reclamation site. The ponds are adjacent to Tehachapi Creek. A storm drain channel, tributary to the Creek, cuts across the WWTF and conveys storm water runoff from approximately 2800 acres north of Highway 58. A 5-foot high dike is at the junction of the drainage channel and the Creek. A 23-acre parcel, west of and adjacent to Ponds 12 to 14, was used from December 1990 to April 1991 for emergency flows. An 8- to 12-foot dike between the Creek and the parcel was constructed to prevent wastewater in the 23-acre parcel from entering the Creek. FEMA maps show that the WWTF is within the 100-year floodway (Zone A). The Discharger completed improvements against a 100-year flood for the WWTF in November 1991.

In the interim prior to completion of the new WWTF, the Discharger will abandon the Imhoff tanks and trickling filter; convert 7 of the existing disposal ponds into 2 aerated ponds and 5 stabilization ponds for treatment; continue using 6 of the existing ponds for disposal; use a 100 acre-feet storage reservoir to store treated wastewater during the winter; and continue reclamation in the 140-acre alfalfa field, as weather permits.

The proposed new 1.25-mgd WWTF will include a control building, headworks, a 1.25-mgd oxidation ditch, a secondary clarifier, about 4000 square feet of sludge drying beds, the 100 acre-feet storage reservoir, and the 140-acre alfalfa reclamation site. The Discharger proposes to use sludge from the WWTF as soil amendment. This Order contains two effluent flow limits. The flow limit of 0.85 represents the interim available treatment and disposal capacity and will be in effect until the Discharger demonstrates that it has treatment and disposal

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capacity of 1.25 mgd. The flow of 1.25 mgd will be in effect after the Discharger demonstrates that it has treatment and disposal capacity of 1.25 mgd.

Effluent from oxidation ponds and biological WWTFs contain significant number of nitrifying organisms which exert an oxygen demand. Without a nitrogen inhibiting chemical, an effluent BOD₅ analysis measures the nitrogenous and carbonaceous oxygen demands. This test, therefore, does not accurately represent the oxygen demand exerted by organic material. Accordingly, these tentative requirements (and the proposed amended Order) allow the Discharger to demonstrate compliance with either BOD₅ or the carbonaceous oxygen demand (CBOD₅). The CBOD₅ represents the oxygen demand exerted by organic organisms.

The Discharger completed preliminary plans and specifications for the new WWTF in 1990, began construction of the headworks in August 1991, and plans to begin construction of the rest of the new WWTF by July 1992 and complete construction by August 1994. The Discharger is in State Water Resources Control Board's State Revolving Fund Program priority list for consideration of a low interest loan to fund the new plant and maximize reclamation.

The reclamation site is adjacent to the local airport. There are several residential developments about 500 feet east of the reclamation area. The storage reservoir is about 2 1/2 miles southeast of the existing WWTF and is surrounded by predominantly agricultural lands. Winds at the airport and the storage reservoir are predominantly towards the east and west. The bottom of the disposal ponds are underlain by approximately 4 to 15 feet of sand/silty sand with a percolation rate of about 0.03 to 0.04 feet per day. Soils at the reclamation area are classified as sandy silts, silty sands, and clays, with permeabilities varying from 9×10^{-9} to 6×10^{-6} (BSK, 1990). Soils beneath the storage reservoir to a depth of 30 feet are silty sands and sandy silts with low to very low permeabilities (Corollo 1991).

Tehachapi Creek is an ephemeral Westside stream and originates in the Tehachapi Valley, about two miles south of the City. At Keene Ranch Tehachapi Creek joins Caliente Creek one-half mile west of the town of Caliente. Caliente Creek drains to the valley floor. The beneficial uses of Tehachapi Creek are agricultural use, preservation of rare and endangered species, wildlife habitat, non-contact water recreation (Keene Ranch), and ground water recharge.

Ground water in the Tehachapi Valley is found at 0 to 300 feet below the ground surface (bgs) and moves generally to the north. Within the City, ground water moves easterly and is about 50 to 80 feet bgs. Data from the City's domestic wells indicate that ground water is of good to excellent quality (EC of 369 to 625 μ mhos/cm, NO₃ of 0 to 4 mg/l, TDS of 300 to 430 mg/l, and pH of 7.8). However, preliminary ground water information collected from the 23-acre parcel in 1990 indicate that ground water beneath this parcel is about 7 to 20 feet bgs and has constituents in concentrations which exceed MCLs for EC, TDS, pH, NO₃, SO₃, Al, and Se. The Discharger will install and develop a ground water monitoring in the area of the ponds and 23-acre parcel to determine the effect of the discharge on ground water quality.

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On 14 June 1991, the Discharger certified a final Environmental Impact Report for the interim improvements and new 1.25-mgd WWTF in accordance with the California Environmental Quality Act (Public Resources Code Section 21000, et seq.) and the State Guidelines. Compliance with these requirements will mitigate and prevent any significant impacts on water quality.

